

Formal charge (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to apply formal charge to determine the preferred Lewis formula when alternative Lewis formulas exist for a species.

As we have established throughout the past sections, the molecular geometry of a molecule is important. It is useful in predicting a molecule's structural properties, such as polarity, melting and boiling points, and so on, and can affect how the molecule will react. The shape of a molecule is dependent on the number of electron domains, which can be identified based on the Lewis formula of that molecule.

Consider a molecule of carbon dioxide, CO₂. Use [section S2.2.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>) to help you suggest the Lewis formula for this molecule.

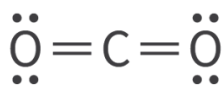


Figure 1. Lewis formulas of carbon dioxide.

 More information for figure 1

The image shows the Lewis structure for the carbon dioxide molecule. At the center, there is a letter 'C' representing the carbon atom. On either side of the carbon atom, there is a letter 'O' representing oxygen atoms. Lines are drawn between the carbon atom and each oxygen atom to indicate double bonds. Each oxygen atom has four dots around it, representing the lone pairs of electrons not involved in bonding. The structure visually represents the arrangement of atoms and electrons in a carbon dioxide molecule.

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You may have identified that carbon is the central atom and drawn a Lewis formula similar to the one in **Figure 1**. But, how can we be sure that this is correct?

In theory, another Lewis formula could also be drawn for the same molecule, which is shown in **Figure 2**.



Figure 2. Potential alternative Lewis formulas of CO₂.

 More information for figure 2

The image depicts a Lewis structure diagram of carbon dioxide (CO_2). In this representation, the carbon atom (C) is centered. On the left, it is triple-bonded to an oxygen atom (O), represented by three parallel lines. This oxygen atom has four dots around it, indicating lone pairs of electrons. On the right, the carbon atom is single-bonded to another oxygen atom, shown by a single line. This oxygen atom has six dots around it, representing the three lone pairs of electrons present. This alternative Lewis formula adds up to the required 16 valence electrons and complies with the octet rule.

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In theory, both these are valid structures: they fulfil the octet rule and contain the 16 valence electrons that must be present. How can we verify that it is the first molecule that actually exists?

The need for formal charge

As you can see from the CO_2 example, more than one valid Lewis formulas can theoretically exist for certain species and is particularly common for molecules that can form an expanded octet. In reality, one of the structures will always be far more likely to actually exist. Just because a structure fits the rules for drawing Lewis formulas, does not mean it is right!

It is formal charge that allows us to determine which one is more likely. Formal charge is a method of electron 'bookkeeping', it is a simple way to compare the number of valence electrons an atom has around it before and after bonding.

Making connections

Section S2.2.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/>) covers the concept of electronegativity and how it determines the polarity of bonds, and contributes to overall molecular polarity (when combined with the symmetry of the molecule).

Section S3.1.3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/>) covers more about electronegativity, including its trend in the periodic table.

Although you may already be aware that this is not true; formal charge assumes all atoms as if they have identical electronegativity and all bonds as if they have pure covalent character with equal electron distribution.

Formal charge is based on counting how many electrons 'belong' to each atom in a Lewis formula, and comparing this with the number of valence electrons in the non-bonded atom. The difference between these two values is the formal charge.

When this is done for several Lewis formulas, it is possible to establish the most likely structure to exist; it will always be the one with the formal charge closest to zero for as many atoms in the structure as possible. This is because a low formal charge means that less charge transfer has taken place in forming the structure, indicating that the number of valence electrons before and after bonding is similar. This results in a more stable structure overall.

Concept

Note that resonance is used to explain several Lewis formulas that differ in terms of electron distribution, that is, in molecules where there is potential for bonds to be placed in different positions.

Formal charge is used to determine the most likely Lewis formula when many structures exist that differ in terms of atom arrangement or electron distribution.

How to calculate formal charge

Formal charge is calculated by subtracting the non-bonding and bonded pairs of electrons from the original valence pairs of valence electrons for an atom before bonding has taken place (**Figure 3**).

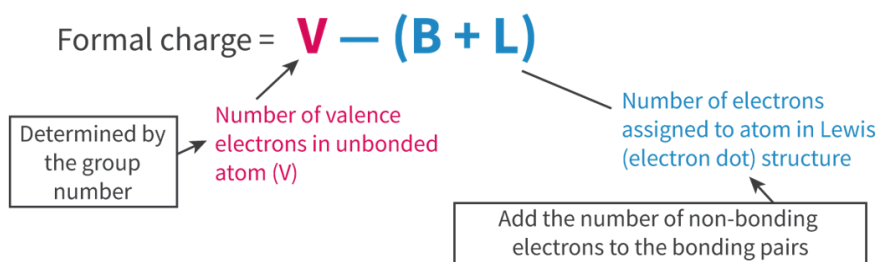


Figure 3. How to calculate formal charge.

 More information for figure 3

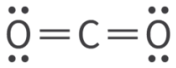
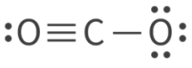
The image presents the equation for calculating the formal charge of an atom in a molecule, accompanied by annotations. The formula is: Formal charge = $V - (B + L)$. Here, "V" represents the number of valence electrons in an unbonded atom, labeled as "Number of valence electrons in unbonded atom (V)," pointing to "V" in the formula. This is determined by the group number, as noted in a box expressing "Determined by the group number." "B" and "L" together represent the number of electrons around the atom assigned in the Lewis structure. "B" refers to the non-bonding electrons and "L" to the bonding pairs, shown with the label "Number of electrons assigned to atom in Lewis (electron dot) structure." Another box indicates "Add the number of non-bonding electrons to the bonding pairs," explaining how to count these electrons. The overall layout illustrates how the different components are used to calculate formal charge, with directional arrows pointing to each part of the formula.

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At the beginning of this section, we represented a molecule of CO_2 using two different Lewis formulas. We can use formal charge to identify which of these is the most likely structure.

Worked example 1

Two versions of the CO_2 molecule exist: use formal charge to identify the most likely structure.

Structure A	Structure B
 More information The image is a diagram showing the structure of a carbon dioxide (CO₂) molecule. It features a central carbon atom "C" bonded to two oxygen atoms "O," one on each side. The bonds are double bonds, represented by two parallel lines between each oxygen and the central carbon. Each oxygen atom has four additional dots around it, indicating four lone pairs of electrons. The diagram illustrates the linear configuration of the molecule. [Generated by AI]	 More information The image shows the chemical structure diagram of carbon dioxide (CO₂). It features a linear arrangement with three main components: a central carbon atom in the center, double-bonded to two oxygen atoms on either side. The double bonds are represented by two parallel lines between oxygen and the central carbon. Each oxygen atom has two pairs of dots representing lone pairs around them, indicated by dots placed near oxygen atoms. [Generated by AI]

	S
Carbon	V = 4 (C is in group 14) 4 bonding pairs 0 lone electrons $FC = 4 - (4 + 0)$ FC = 0
Oxygen	O = 6 (O is in group 16) 2 bonding pairs 4 lone electrons $FC = 6 - (2 + 4)$ FC = 0
	S
Carbon	V = 4 (C is in group 14) 4 bonding pairs 0 lone electrons $FC = 4 - (4 + 0)$ FC = 0
Oxygen (left)	O = 6 (O is in group 16) 3 bonding pairs 2 lone electrons

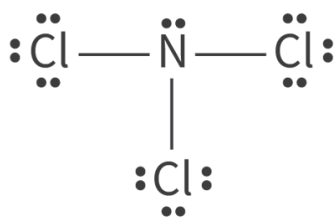
	S
	$FC = 6 - (3 + 2)$ $FC = +1$
Oxygen (right)	$O = 6$ (O is in group 16) 1 bonding pairs 6 lone electrons $FC = 6 - (1 + 6)$ $FC = -1$

As you can see, in Structure A the formal charge on the carbon and both oxygens is 0, indicating a stable structure. Compare this to Structure B: carbon still has a formal charge of 0, but both oxygens have a charge.

As the one with the formal charge closest to zero for as many atoms in the structure as possible, Structure A is the most likely structure.

Worked example 2

Calculate the formal charges on the nitrogen trichloride molecule, NCl_3



More information

The image displays the Lewis structure of the nitrogen trichloride (NCl_3) molecule. At the center, there is a nitrogen atom represented by the letter 'N' with a pair of dots above it indicating a lone pair of electrons. Surrounding the nitrogen are three chlorine atoms represented by the letters 'Cl'. Two chlorine atoms (the left and the right one) are linked to the nitrogen atom via single lines, symbolizing single covalent bonds. Each chlorine atom has three pairs of dots around it depicting their lone pairs of electrons. The structure is arranged in a T-shape with one chlorine atom below the central nitrogen.

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Element	Formal Ch
Nitrogen	$V = 5$ (N is 3 bonding 2 lone elec

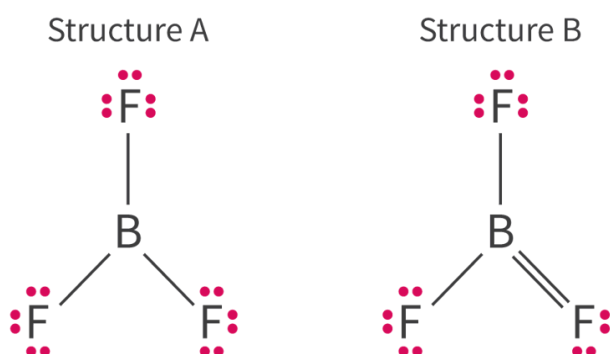
Element	Formal Ch
	$FC = 5 - ($ $FC = 0$
Chlorine (all chlorines have the same electron arrangement around them so no need to calculate them separately)	$V = 7 \text{ (Cl is 1 bonding, 6 lone elec)}$ $FC = 7 - ($ $FC = 0$

Formal charge on nitrogen and all chlorines is 0, indicating a stable structure.

We can also apply this concept to better understand why a molecule exists with an incomplete octet, such as BF_3 .

Worked example 3

Calculate the formal charges on the boron trifluoride molecules, BF_3 , to identify which is the most likely structure.



More information

The image shows two Lewis structures of boron trifluoride molecules labeled "Structure A" and "Structure B." Both structures feature a central boron (B) atom bonded to three fluorine (F) atoms. In Structure A, each of the three fluorine atoms is singly bonded to the boron atom, and each fluorine is surrounded by three pairs of dots representing six valence electrons. Structure B additionally illustrates a double bond between one of the fluorine atoms and the boron atom, while the others remain singly bonded. Each fluorine atom still retains six non-bonding electrons represented by three pairs of dots.

[Generated by AI]

Structure A	Structure B

Boron	$V = 3$ (B is in group 13) 3 bonding pairs 0 lone electrons $FC = 3 - (3 + 0)$ $FC = 0$
Fluorine	$V = 7$ (F is in group 17) 1 bonding pairs 6 lone electrons $FC = 7 - (1 + 6)$ $FC = 0$

Boron	$V = 3$ (B is in group 13) 4 bonding pairs 0 lone electrons $FC = 3 - (4 + 0)$ $FC = -1$
Fluorine (top and bottom left)	$V = 7$ (F is in group 17) 1 bonding pairs 6 lone electrons $FC = 7 - (1 + 6)$ $FC = 0$
Fluorine (bottom right)	$V = 7$ (F is in group 17) 2 bonding pairs 4 lone electrons $FC = 7 - (2 + 4)$ $FC = +1$

Many of the atoms within Structure B have a formal charge that is not equal to 0 and can be reduced, whereas those in Structure A are all 0. Therefore, Structure A is the preferred structure.

We can also apply this concept of formal charge to identify the most likely structure of hydrogen cyanide, HCN, which theoretically could have either the C or N placed as the central atom. The rules we identified in [section S2.2.4](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>) indicate that if carbon is present, then it should be the central atom and that the atom which is furthest left is the central atom. Can we use formal charge to prove this?

	H—C≡N:	H—N≡C:
H	V = 1 (H is in group 1) 1 bonding pair 0 lone electrons $FC = 1 - (1 + 0)$ $FC = 0$	V = 1 (H is in group 1) 1 bonding pair 0 lone electrons $FC = 1 - (1 + 0)$ $FC = 0$
C	V = 4 (C is in group 14) 4 bonding pairs 0 lone electrons $FC = 4 - (4 + 0)$ $FC = 0$	V = 4 (C is in group 14) 3 bonding pairs 2 lone electrons $FC = 4 - (3 + 2)$ $FC = -1$
N	V = 5 (N is in group 15) 3 bonding pairs 2 lone electrons $FC = 5 - (3 + 2)$ $FC = 0$	V = 5 (N is in group 15) 4 bonding pairs 0 lone electrons $FC = 5 - (4 + 0)$ $FC = +1$

The structure that has carbon as the central atom is therefore the most likely structure as the structure has the lowest formal charge. This verifies what we learned previously about selecting a central atom.

Worked example 4

Two versions of the SO₂ molecule exist, one where sulfur respects the octet rule and one where it violates it by expanding its octet.

Use formal charge to identify the most likely structure.



 More information

The image shows two chemical structures labeled 'Structure A' and 'Structure B.' Both structures include one sulfur (S) atom bonded with two oxygen (O) atoms. In Structure A, the sulfur atom forms a double bond with the first oxygen atom on the left and a single bond with the second oxygen atom on the right, each oxygen atom having a dash above it indicating a negative charge. In Structure B, the sulfur atom forms double bonds with both oxygen atoms, which similarly have negative charges indicated by dashes above them.

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Step 1: Calculate formal charge for each atom in each structure. The electrons surrounding the oxygens within each molecule are different so they have to be calculated separately.

	Structure A	
Left oxygen	$V = 6$ (O is in group 16) 2 bonding pairs 4 lone electrons $FC = 6 - (2 + 4)$ $FC = 0$	V 2 4 FC FC
Sulfur	$V = 6$ (S is in group 16) 3 bonding pairs 2 lone electrons $FC = 6 - (3 + 2)$ $FC = +1$	V 4 2 FC FC
Right oxygen	$V = 6$ (O is in group 16) 1 bonding pairs 6 lone electrons $FC = 6 - (1 + 6)$ $FC = -1$	V 2 4 FC FC

Step 2: Identify the most likely structure. Structure B with the expanded octet is more likely as the formal charge on each atom is 0.

Making connections

So far in this subtopic you have encountered:

- charge (of anions and cations in ionic bonding)
- partial charge (with polar bonds in covalent molecules)
- formal charge (a theoretical, calculated charge).

They are all part of the same subtopic but mean quite different things!

Section S3.1.6 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/>) introduces another charge-related concept: oxidation states.

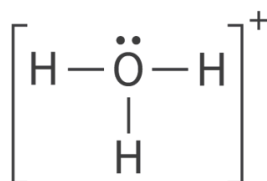
Atoms can gain a formal charge if a coordination bond has been formed. A coordination bond is formed when both electrons being shared are donated by just one of the atoms (see section S2.2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>)).

Because the donor atom has 'given away' an electron, it now has a formal charge of +1 and the receiver atom has a formal charge of -1 as it has 'gained' an electron.

This can be seen in the example with a hydronium ion, H_3O^+ , as follows.

Worked example 5

Calculate the formal charge on each atom in a hydronium ion, H_3O^+ .



 More information

The diagram shows the structural representation of a hydronium ion, H_3O^+ . In the center, an oxygen atom is shown with a formal positive charge, indicated by a plus sign in brackets surrounding the entire structure. It also has two pairs of dots representing lone pairs of electrons. The oxygen is bonded to three hydrogen atoms through single bonds, arranged in a pyramidal shape. The bonds are represented by solid lines. The overall structure is enclosed in brackets with a positive charge symbol, indicating the ion's charge.

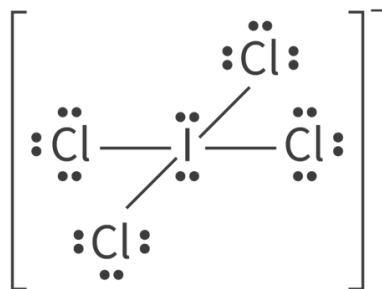
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Element	Formal Charge Calculation
Hydrogens (all)	$V = 1$ (H is in group 1) 1 bonding pairs 0 lone electrons $\text{FC} = 1 - (1 + 0)$ $\text{FC} = 0$
Oxygen	$V = 6$ (O is in group 16) 3 bonding pairs 2 lone electrons $\text{FC} = 6 - (3 + 2)$ $\text{FC} = +1$

We can double check formal charge calculations by determining the sum of the formal charges of all atoms in a molecule or ion and comparing this value to the ionic charge. In a molecular compound, the sum of the formal charges must equal zero, as there are no ionic charges present. In an ion, the sum of the formal charges must equal the charge of the ion, reflecting the number of electrons that have been gained or lost.

Worked example 6

Calculate the formal charges on the ICl_4^- ion below



 More information

The image depicts the Lewis structure of the ICl_4^- ion. In the center, there is an iodine atom (I) surrounded by four chlorine atoms (Cl) at the corners of a square planar shape. Each chlorine atom is depicted with three pairs of dots representing non-bonding electrons. The iodine atom in the center is connected by single lines to each of the chlorine atoms, representing single covalent bonds. Additionally, the iodine atom shows two pairs of dots indicating non-bonding electron pairs. Brackets surround the entire structure with a minus sign (—) outside, denoting the negative charge of the ion.

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Element	Formal Charge Calc
Iodine	$V = 7$ (I is in group 17) 4 bonding pairs 4 lone electrons $FC = 7 - (4 + 4)$ $FC = -1$
All Chlorines	$V = 7$ (Cl is in group 17) 1 bonding pairs 6 lone electrons $FC = 7 - (1 + 6)$ $FC = 0$

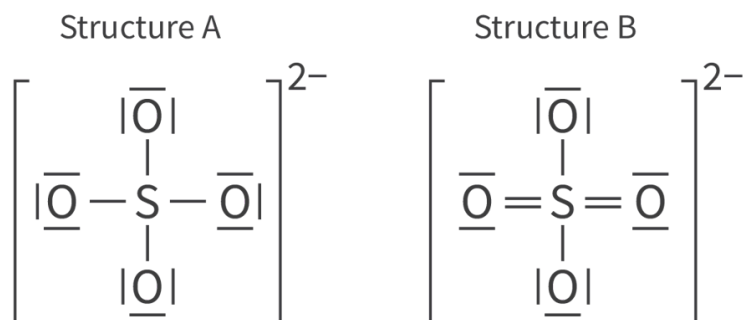
The net formal charge, -1 , is identical to the overall charge of the ion, 1^- , because this ion contains one additional electron.

Remember that not all atoms in the molecule or ion will have a formal charge of zero: the goal is to have *as many as possible* with a formal charge of zero.

Worked example 7

Two versions of the SO_4^{2-} ion exist, one where sulfur respects the octet rule and one where it violates it by expanding its octet.

Use formal charge to identify the most likely structure.



 More information

The image shows two chemical structures labeled as Structure A and Structure B. Both structures are enclosed within square brackets with a 2- charge indicated outside the top right of each bracket.

Structure A: In the center, a sulfur (S) atom is bonded to four oxygen (O) atoms. Each oxygen is shown with a negative sign, indicating it carries a negative charge. The arrangement forms a cross shape, with the sulfur in the center and an oxygen at the top, bottom, left, and right.

Structure B: In the center, a sulfur (S) atom is double bonded to two oxygen (O) atoms on its left and right sides. The double bonds are depicted as two parallel lines. There are also single bonds to two additional oxygen atoms at the top and bottom, each carrying a negative charge, shown with a minus sign above them. The overall symmetry aligns along a vertical axis with the sulfur atom at the center.

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Step 1: Calculate formal charge for each atom in each structure.

	Str
Sulfur	$V = 6$ (S is in group 16) 4 bonding pairs 0 lone electrons $FC = 6 - (4 + 0)$ $FC = +2$
Oxygen (all)	$V = 6$ (O is in group 16) 1 bonding pairs 6 lone electrons

	Str
	$FC = 6 - (6 + 1)$ $FC = -1$
	Str
Sulfur	$V = 6$ (S is in group 16) 6 bonding pairs 0 lone electrons $FC = 6 - (6 + 0)$ $FC = 0$
Oxygen (top and bottom)	$V = 6$ (O is in group 16) 1 bonding pairs 6 lone electrons $FC = 6 - (6 + 1)$ $FC = -1$
Oxygen (left and right)	$V = 6$ (O is in group 16) 2 bonding pairs 4 lone electrons $FC = 6 - (4 + 2)$ $FC = 0$

Step 2: Identify the most likely structure.

Structure A will have a net charge of -2

$$(1 \times S) + (4 \times O)$$

$$(+2) + (4 \times -1) = -2 \text{ overall}$$

Structure B will have a net charge of -2

$$(1 \times S) + (2 \times O) + (2 \times O)$$

$$(0) + (2 \times -1) + (2 \times 0) = -2 \text{ overall}$$

Both structures have a net formal charge that is the same as the charge on the ion.

Structure B with the expanded octet is more likely as it has more atoms in the molecule with a formal charge of 0.

Limitations of formal charge

Formal charge is a useful tool in helping us identify the preferred structure of a molecule or ion when two or more alternatives exist. However, one of its limitations is that it does not consider the electronegativity of the atoms in the molecule.

Take dinitrogen oxide, N_2O , for example. Two valid Lewis formulas for N_2O can be drawn (**Figure 4**).



Figure 4. Two potential Lewis formulas for dinitrogen oxide molecule.

 More information for figure 4

The image displays two potential Lewis structures for the dinitrogen oxide (N_2O) molecule. Structure A shows a nitrogen atom on the left with a triple bond to another nitrogen atom, which is single-bonded to an oxygen atom. The nitrogen and oxygen atoms have lone pairs of electrons. Structure B depicts a nitrogen atom with a lone pair of electrons, bonded by a double bond to another nitrogen atom, which is double-bonded to an oxygen atom with a lone pair. The image illustrates the electron arrangement in dinitrogen oxide using Lewis dot structures.

[Generated by AI]

Formal charges can be assigned that would result in the molecule shown in **Figure 5**.

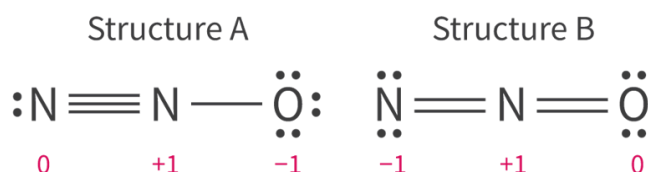


Figure 5. Two potential Lewis formulas for dinitrogen oxide molecule, showing formal charges.

 More information for figure 5

The image displays two Lewis structures for a dinitrogen oxide molecule. Both structures are labeled as Structure A and Structure B. In Structure A, the molecule is depicted as a nitrogen atom connected by a triple bond to another nitrogen, which is then connected by a single bond to an oxygen atom. The formal charges are shown below each atom: the first nitrogen has a charge of 0, the second nitrogen has a charge of +1, and the oxygen has a charge of -1. In Structure B, the first nitrogen is bonded by a double bond to the second nitrogen, which is also double-bonded to an oxygen atom. The formal charges here are: the first nitrogen has a charge of -1, the second nitrogen has a charge of +1, and the oxygen has a charge of 0. The presence of bonding electrons and lone pairs are shown with dots.

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Both have the same formal charge on atoms and the same net charge, which would suggest equal stability.

However, Structure A is actually the observed structure as the negative charge is found on the more electronegative oxygen atom, which is a more realistic reflection of the electron distribution in that molecule.

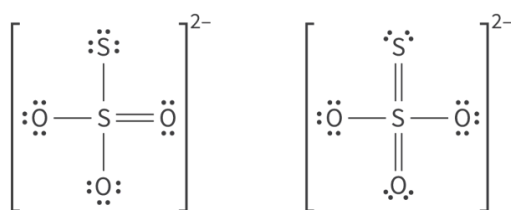
This activity is designed to allow you to apply the knowledge covered in this section about formal charge and how we can use it to determine the most likely structure to exist when more than one valid Lewis formulas can be drawn for a molecule.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 25 minutes
- **Activity type:** Individual activity

1. For each of the following pairs of structures, determine the formal charge on each atom and determine the most plausible Lewis formula.

(a)

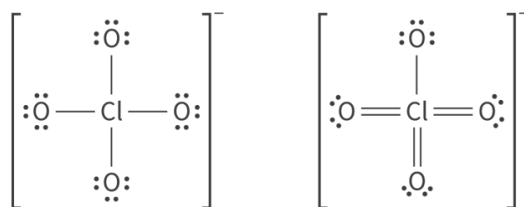


 More information

The image features two Lewis structures of sulfur dioxide with a -2 charge, both within brackets to indicate charged species. The left structure shows a central sulfur atom double-bonded to an oxygen atom at the top and a single-bonded oxygen atom on the left, with lone pairs visible. To the right, another oxygen is double-bonded, also showing lone pairs. The top sulfur exhibits a lone pair above it. The right side structure reverses the double bonds: one between sulfur and the top oxygen, another between sulfur and the bottom oxygen, both showing lone pairs, while the sulfur bonds singly to the oxygens on either side, with lone pairs present. The sulfurs in each center have formal charges indicating their oxidation states, with visual emphasis on the bonds and electron pairs to determine plausibility of the structures according to formal charge rules.

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(b)



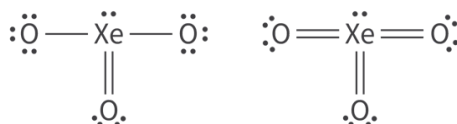
More information

The image displays two chemical structure diagrams of chlorine oxoanions. On the left, the diagram shows a chlorate ion with a central chlorine atom bonded to three oxygen atoms. Each oxygen atom is depicted with its valence electrons represented as dots. The overall charge of the ion is negative, indicated by a bracket and a minus sign.

On the right, the diagram illustrates a different chlorine oxoanion with a central chlorine atom double-bonded to one oxygen atom and single-bonded to two other oxygen atoms. The lone pairs of electrons on the oxygen atoms are shown as dots. This ion is also negatively charged, shown by a bracket and minus sign. The structures are presented side by side.

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(c)



More information

The image consists of two chemical structural diagrams of Xenon trioxide (XeO_3) molecules. Each diagram displays a central Xenon (Xe) atom with three Oxygen (O) atoms bonded to it. The diagram on the left shows the central Xe atom with single bonds to each of the three O atoms. Each Oxygen has two lone pairs symbolized by dots. The Xe also displays a lone pair of electrons. The diagram on the right is similar but indicates a different arrangement or resonance structure with double bonds between the Xe and O atoms. Such diagrams illustrate the possible Lewis structures representing the molecule's electron arrangements and bonding. The images help in understanding the molecular geometry and electron distribution around the Xenon atom in Xenon trioxide.

[Generated by AI]

- Potassium bromate is often added to flour used for baking bread as it results in strong dough which can rise higher due to additional elasticity which KBrO_3 provides.

There are two possible Lewis formulas for the bromate ion, BrO_3^- .

Draw both these structures and determine the most plausible structure based on their formal charge.